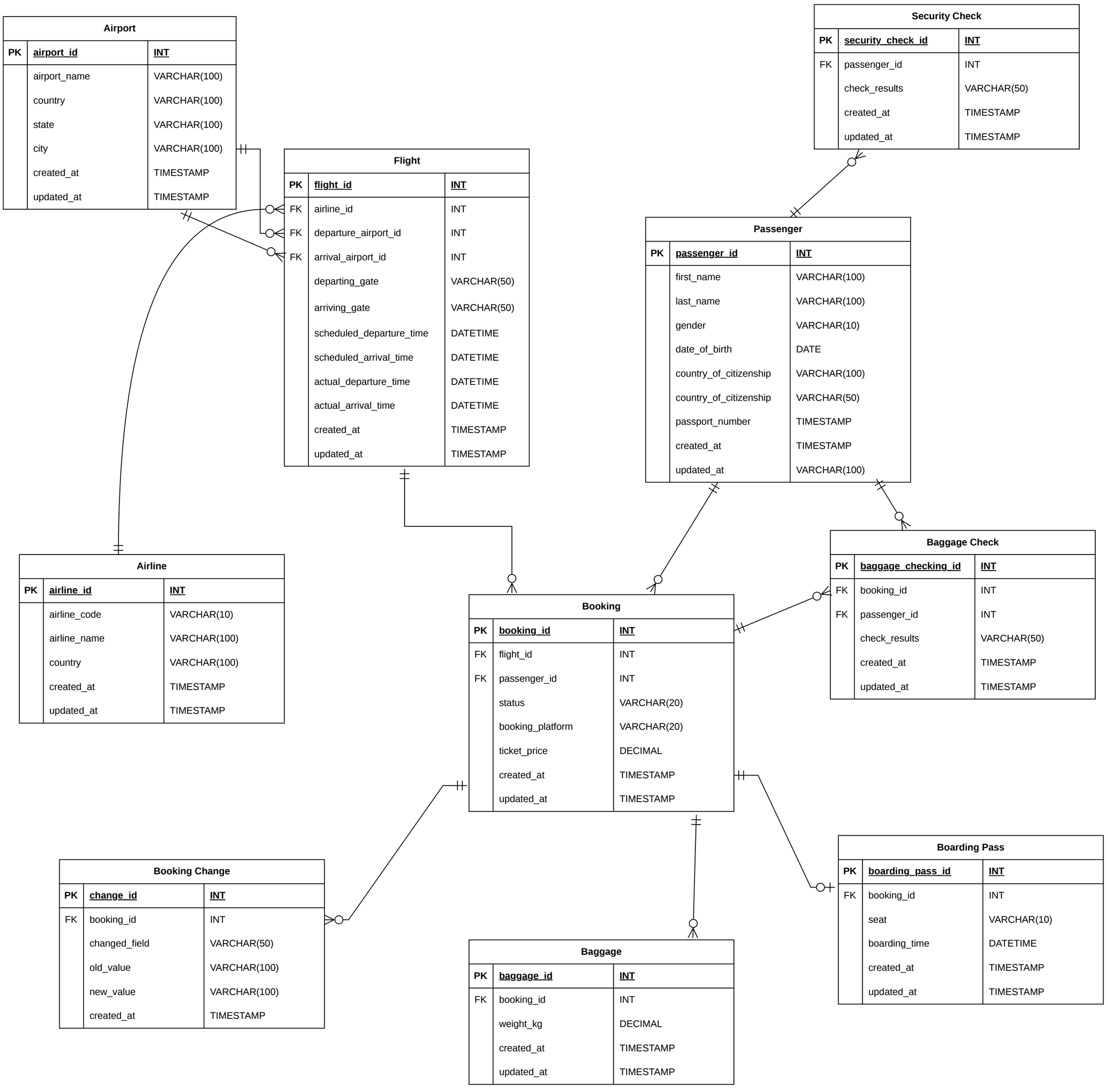

Airport
PK airport_id INT
airport_name VARCHAR(100)
country VARCHAR(100)
state VARCHAR(100)
city VARCHAR(100)
created_at TIMESTAMP
updated_at TIMESTAMP
Flight
PK flight_id INT
FK airline_id INT
FK departure_airport_id INT
FK arrival_airport_id INT
departing_gate VARCHAR(50)
arriving_gate VARCHAR(50)
scheduled_departure_time DATETIME
scheduled_arrival_time DATETIME
actual_departure_time DATETIME
actual_arrival_time DATETIME
created_at TIMESTAMP
updated_at TIMESTAMP
Security Check
PK security_check_id INT
FK passenger_id INT
check_results VARCHAR(50)
created_at TIMESTAMP
updated_at TIMESTAMP
Passenger
PK passenger_id INT
first_name VARCHAR(100)
last_name VARCHAR(100)
gender VARCHAR(10)
date_of_birth DATE
country_of_citizenship VARCHAR(100)
country_of_citizenship VARCHAR(50)
passport_number TIMESTAMP
created_at TIMESTAMP
updated_at VARCHAR(100)
Airline
PK airline_id INT
airline_code VARCHAR(10)
airline_name VARCHAR(100)
country VARCHAR(100)
created_at TIMESTAMP
updated_at TIMESTAMP
Booking
PK booking_id INT
FK flight_id INT
FK passenger_id INT
status VARCHAR(20)
booking_platform VARCHAR(20)
ticket_price DECIMAL
created_at TIMESTAMP
updated_at TIMESTAMP
Baggage Check
PK baggage_checking_id INT
FK booking_id INT
FK passenger_id INT
check_results VARCHAR(50)
created_at TIMESTAMP
updated_at TIMESTAMP
Booking Change
PK change_id INT
FK booking_id INT
changed_field VARCHAR(50)
old_value VARCHAR(100)
new_value VARCHAR(100)
created_at TIMESTAMP
Baggage
PK baggage_id INT
FK booking_id INT
weight_kg DECIMAL
created_at TIMESTAMP
updated_at TIMESTAMP
Boarding Pass
PK boarding_pass_id INT
FK booking_id INT
seat VARCHAR(10)
boarding_time DATETIME
created_at TIMESTAMP
updated_at TIMESTAMP

Entities:

- **Airport** - stores information about each airport.
- **Airline** - stores details of airline companies operating flights.
- **Flight** - records scheduled and actual flight details.
- **Passenger** - personal informationof passengers.
- **Booking** - passenger's reservation on a specific flight.
- **BookingChange** - records the history ofany modifications made to a booking.
- **BoardingPass** - stores the boarding pass issued for a booking.
- **Baggage** - baggage details registered for a booking.
- **BaggageCheck** - records the results of baggage screening for a booking and passenger.
- **SecurityCheck** - records the results of airport security screening for a passenger.

Relationships:

Almost all relationships are optional-one-to-mandatory-many: the "one" side can exist in the database without any related records yet, while the "many" side must always be linked to exactly one parent record to exist at all.

- Airport → Flight (departure): one airport optionally has many flights; each flight must have exactly one departure airport.
- Airport → Flight (arrival): one airport optionally has many flights; each flight must have exactly one arrival airport.
- Airline → Flight: one airline optionally has many flights; each flight must belong to exactly one airline.
- Flight → Booking: one flight optionally has many bookings; each booking must belong to exactly one flight.
- Passenger → Booking: one passenger optionally has many bookings; each booking must belong to exactly one passenger.
- Booking → BookingChange: one booking optionally has many change records; each change record must belong to exactly one booking.
- Booking → Baggage: one booking optionally has many baggage items (only if the passenger has baggage); each baggage record must belong to exactly one booking.
- Booking → BaggageCheck: one booking optionally has many baggage checks; each baggage check must belong to exactly one booking.
- Passenger → BaggageCheck: one passenger optionally has many baggage checks; each baggage check must belong to exactly one passenger.
- Passenger → SecurityCheck: one passenger optionally has many security checks; each security check must belong to exactly one passenger.

Except Booking ↔BoardingPass, which is one-to-one: each booking can optionally have at most one boarding pass, while each boarding pass belongs to exactly one booking.

Notation key:

- **PK** — Primary Key
- **FK** — Foreign Key